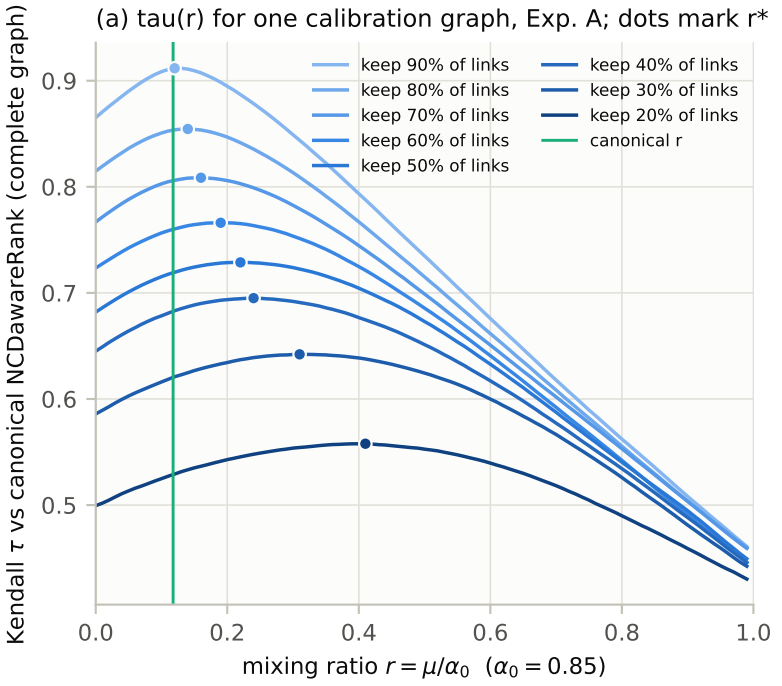
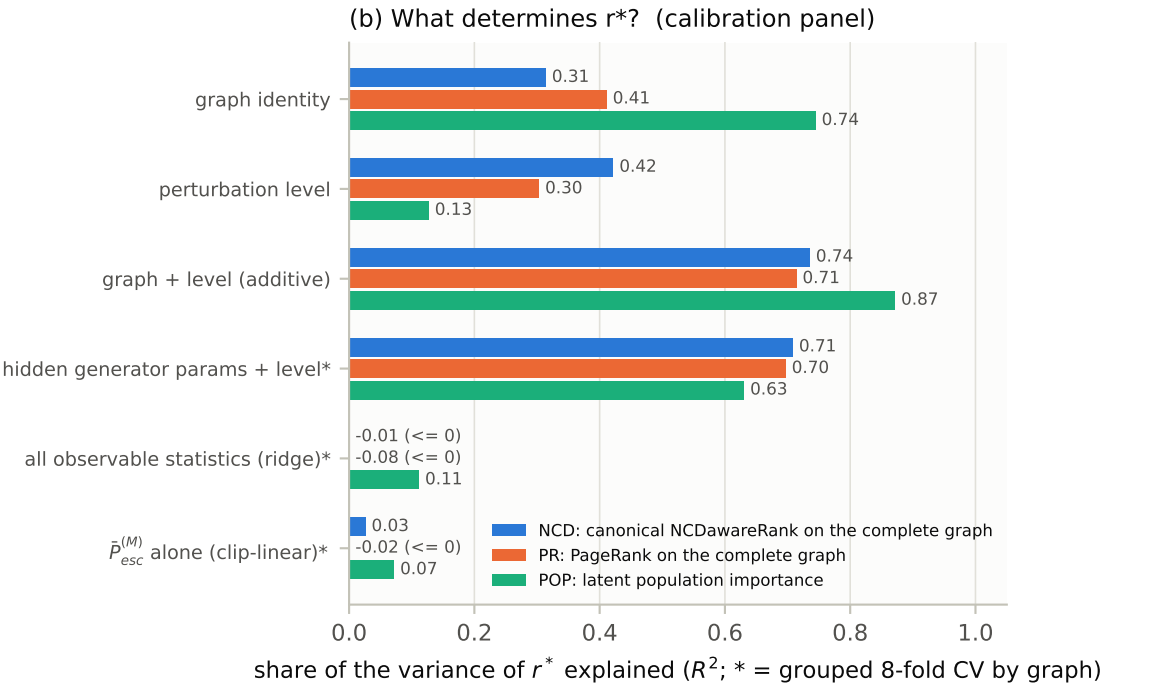
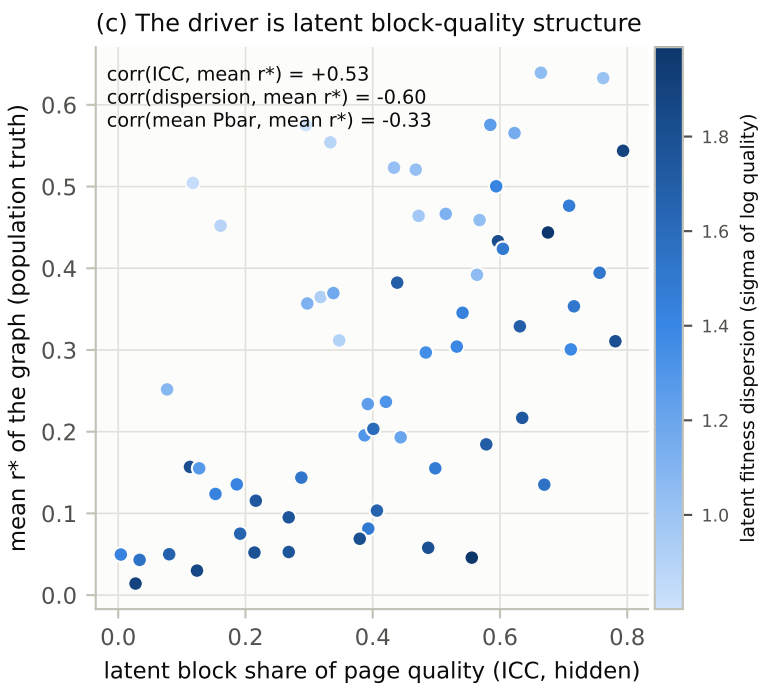
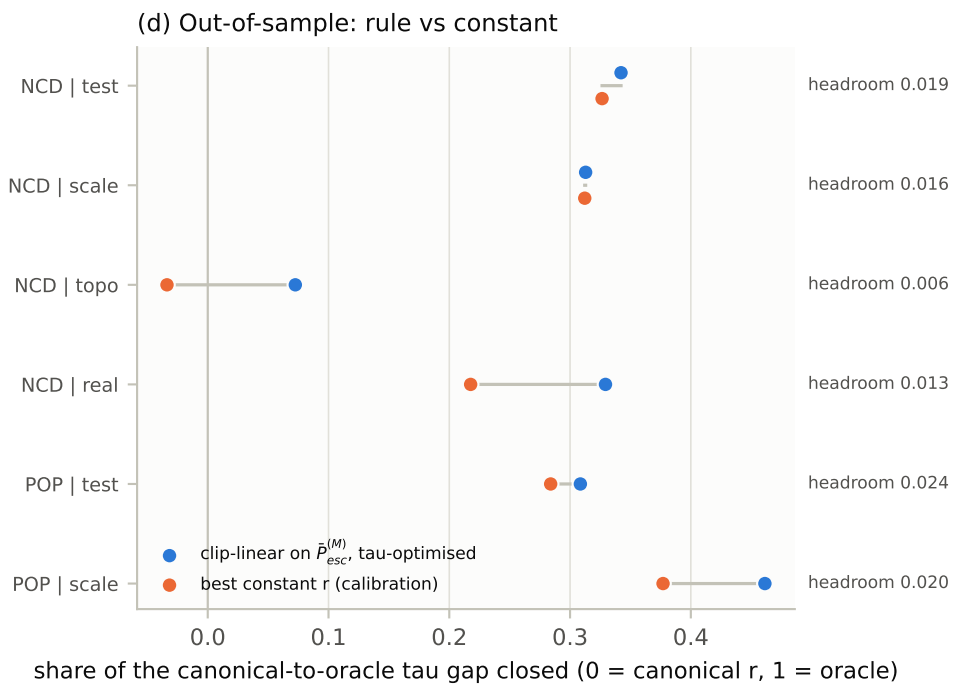

(a) tau(r) for one calibration graph, Exp. A; dots mark r*
keep 90% of links
keep 80% of links
keep 70% of links
keep 60% of links
keep 50% of links
keep 40% of links
keep 30% of links
keep 20% of links
canonical r
Kendall τ vs canonical NCDawareRank (complete graph)
mixing ratio $r = \mu/\alpha_0$ ($\alpha_0 = 0.85$)


(b) What determines r*? (calibration panel)
graph identity
perturbation level
graph + level (additive)
hidden generator params + level*
all observable statistics (ridge)*
$\tilde{P}^{(M)}_{esc}$ alone (clip-linear)*
0.31
0.41
0.74
0.42
0.30
0.13
0.74
0.71
0.87
0.71
0.70
0.63
-0.01 (<= 0)
-0.08 (<= 0)
0.11
0.03
-0.02 (<= 0)
0.07
NCD: canonical NCDawareRank on the complete graph
PR: PageRank on the complete graph
POP: latent population importance
share of the variance of r^* explained (R^2; * = grouped 8-fold CV by graph)


(c) The driver is latent block-quality structure
corr(ICC, mean r*) = +0.53
corr(dispersion, mean r*) = -0.60
corr(mean Pbar, mean r*) = -0.33
mean r* of the graph (population truth)
latent block share of page quality (ICC, hidden)
latent fitness dispersion (sigma of log quality)


(d) Out-of-sample: rule vs constant
NCD | test
NCD | scale
NCD | topo
NCD | real
POP | test
POP | scale
headroom 0.019
headroom 0.016
headroom 0.006
headroom 0.013
headroom 0.024
headroom 0.020
clip-linear on $\tilde{P}^{(M)}_{esc}$, tau-optimised
best constant r (calibration)
share of the canonical-to-oracle tau gap closed (0 = canonical r, 1 = oracle)